

DU Mathsoc Problem Solving: Set 4

Date: 07/10/2026

~Advanced Paper~

For any assistance or inquiries about the problem sets, please email me at: quizmaster.maths@csc.tcd.ie.

Problem 1. Each edge of a complete graph on $2k$ vertices is removed with probability $\frac{1}{2}$. Prove that with probability greater than $\frac{1}{4^k}$, the maximum degree of the remaining graph is at most $k - 1$.

Problem 2. Suppose a finite number of integer arithmetic progressions are given, such that every positive integer belongs to exactly one of them. Prove that two of these progressions have the same difference.

Problem 3. Do there exist positive integers a and b with $b - a > 1$ such that for every $a < k < b$, either $\gcd(a, k) > 1$ or $\gcd(b, k) > 1$?

Problem 4. Suppose that a finite group has exactly n elements of order p , where p is a prime. Prove that either $n = 0$ or p divides $n + 1$.

Problem 5. If A and B are square matrices of the same size and $ABAB = 0$, does it follow that $BABA = 0$?